L-10

21/08/25

Inverse Laplace transform

(L.T) → time domain → frequency domain

(I.L.T) → frequency domain → time domain

Type-I (Linearity)

$$\mathcal{L}^{-1}\{F(s)+G(s)\} = \mathcal{L}^{-1}\{F(s)\} + \mathcal{L}^{-1}\{G(s)\}$$

$$\mathcal{L}^{-1}\{K\cdot F(s)\} = K\,\mathcal{L}^{-1}\{F(s)\}.$$

① $\mathcal{L}\{1\} = \frac{1}{s}$; $\mathcal{L}^{-1}\{\frac{1}{s}\} = 1$

② $\mathcal{L}\{t^n\} = \frac{n!}{s^{n+1}}$, $\mathcal{L}^{-1}\{\frac{n!}{s^{n+1}}\} = t^n$

③ $\mathcal{L}\{\sin at\} = \frac{a}{s^2+a^2}$; $\mathcal{L}^{-1}\{\frac{a}{s^2+a^2}\} = \sin at$.

e.g $= \frac{2}{3}\mathcal{L}^{-1}\{\frac{3}{s^2+3^2}\} = \frac{2}{3}\sin 3t$

④ $\mathcal{L}^{-1}\{\frac{s}{s^2+a^2}\} = \cos at$.

⑤ $\mathcal{L}\{e^{at}\} = \frac{1}{s-a}$; $\mathcal{L}^{-1}\{\frac{1}{s-a}\} = e^{at}$

Ex 1: Evaluate $\mathcal{L}^{-1}\{1/s^4\}$

$\Rightarrow \mathcal{L}^{-1}\{\frac{1}{s^4}\} = \mathcal{L}^{-1}\{\frac{1}{s^{3+1}}\} = \frac{1}{3!}\mathcal{L}^{-1}\{\frac{3!}{s^{3+1}}\} = \frac{1}{3!}t^3$. Ans:

Ex-2: $\mathcal{L}^{-1}\{\frac{12}{4-3s}\}$.

$= 12\,\mathcal{L}^{-1}\{\frac{1}{-3(s-4/3)}\}. = \frac{-12}{3}\mathcal{L}^{-1}\{\frac{1}{s-4/3}\} = -4e^{4/3t}$ Ans:

Ex-3: $\mathcal{L}^{-1}\{\frac{23s-15}{s^2+4}\} = \mathcal{L}^{-1}\{\frac{23s}{s^2+4} - \frac{15}{s^2+4}\} = \mathcal{L}^{-1}\{\frac{23s}{s^2+4}\} - \mathcal{L}^{-1}\{\frac{15}{s^2+4}\}$.

$= 23\mathcal{L}^{-1}\{\frac{s}{s^2+2^2}\} - \cancel{15\,\mathcal{L}^{-1}\{\frac{1}{s^2-(-4)}\}}\ \frac{15}{2}\mathcal{L}^{-1}\{\frac{2}{s^2+2^2}\}$

$= 23\cos 2t - \frac{15}{2}\sin 2t$ Ans:

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(H.W) $\mathcal{L}^{-1}\left\{\frac{2s-5}{s^2-9}\right\} = ?$

shift from the prev position

Type-II (First Translation th^m:)

$\mathcal{L}^{-1}\{F(s)\} = e^{at}\mathcal{L}^{-1}\{F(s+a)\}$

Step 1 → $F(s+a)$
Step 2 → $\mathcal{L}^{-1}\{F(s+a)\}$
Step 3 → multiply with e^{at}.

$\frac{1}{(s-2)^2+4}$

a → the value which will help to return to the original position.

here, $s = s+2$ → a

$\therefore \frac{1}{(s-2)^2+4} = \frac{1}{(s+2-2)^2+4}$

$a = 2$

Ex-1: $\mathcal{L}^{-1}\left\{\frac{6}{(s-2)^6}\right\}$

Sol^n: $\therefore F(s) = \frac{6}{(s-2)^6}$

$\therefore F(s+a) = F(s+2) = \frac{6}{(s+2-2)^6} = \frac{6}{s^6}$

$\therefore \mathcal{L}^{-1}\{F(s+a)\} = \mathcal{L}^{-1}\{F(s+2)\} = \mathcal{L}^{-1}\left\{\frac{6}{s^6}\right\} = 6\mathcal{L}^{-1}\left\{\frac{1}{s^{5+1}}\right\}$

$= \frac{6}{5!}\mathcal{L}^{-1}\left\{\frac{5!}{s^{5+1}}\right\} = \frac{6}{5!}t^5.$

$\therefore \mathcal{L}^{-1}\left\{\frac{6}{(s-2)^6}\right\} = e^{2t}\cdot\frac{6}{5!}t^5.$ Ans:

Ex-2: $\mathcal{L}^{-1}\left\{\frac{6s-4}{s^2-8s-9}\right\} = \mathcal{L}^{-1}\left\{\frac{6s-4}{s^2-2\cdot4\cdot s+4^2-4^2-9}\right\} = \mathcal{L}^{-1}\left\{\frac{6s-4}{(s-4)^2-25}\right\}$

$\therefore F(s) = \frac{6s-4}{(s-4)^2-25}$; $F(s+a) = \cancel{\frac{6(s+4)}{\ldots}}$

$= F(s+4) = \frac{6(s+4)-4}{(s+4-4)^2-25}$

$= \frac{6s+20}{s^2-25}$

$\therefore \mathcal{L}^{-1}\{F(s+a)\} = \mathcal{L}^{-1}\{F(s+4)\} = \mathcal{L}^{-1}\left\{\frac{6s+20}{s^2-25}\right\}$

$= \mathcal{L}^{-1}\left\{\frac{6s}{s^2-25}\right\} + \mathcal{L}^{-1}\left\{\frac{20}{s^2-25}\right\}$

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$$= 6\mathcal{L}^{-1}\left\{\frac{s}{s^2-25}\right\} + 20\,\mathcal{L}^{-1}\left\{\frac{1}{s^2-25}\right\}$$

$$= 6\mathcal{L}^{-1}\left\{\frac{s}{s^2-25}\right\} + \frac{20}{5}\,\mathcal{L}^{-1}\left\{\frac{5}{s^2-25}\right\}.$$

$$\mathcal{L}^{-1}\left\{\frac{a}{s^2-a^2}\right\} = \sinh at$$

$$\mathcal{L}^{-1}\left\{\frac{s}{s^2-a^2}\right\} = \cosh at$$

$$= 6\cosh 5t + \frac{20}{5}\sinh 5t.$$

$$= 6\cosh(5t) + 4\sinh(5t).$$ Ans:

$$\therefore \mathcal{L}^{-1}\left\{\frac{6s-4}{s^2-8s-9}\right\} = e^{4t}\left(6\cosh(5t) + 4\sinh(5t)\right).$$ Ans:

Type III (2nd Translation th^m)

$$\mathcal{L}^{-1}\{F(s)\cdot e^{-as}\} = \underbrace{f(t-a)}_{2}\cdot u(t-a).$$ 3rd

where, $\mathcal{L}^{-1}\{F(s)\} = f(t)$. $\rightarrow$ 1st step

Ex-1: $\mathcal{L}^{-1}\left\{\frac{1}{s-3}e^{-2s}\right\} = ?$

Sol^n: $F(s) = \frac{1}{s-3}$; $\mathcal{L}^{-1}\{F(s)\} = \mathcal{L}^{-1}\left\{\frac{1}{s-3}\right\} = e^{3t} = f(t)$.

$\therefore f(t) = e^{3t}$; $f(t-a) = f(t-2) = e^{3(t-2)} = e^{3t-6} = e^{3t}\cdot e^{-6}$

$$\therefore \mathcal{L}^{-1}\left\{\frac{1}{s-3}e^{-2s}\right\} = f(t-2)\cdot u(t-2)$$

$$= e^{3t}\cdot e^{-6}\cdot u(t-2)$$

Ans:

Ex-2: $\mathcal{L}^{-1}\left\{\frac{6s-4}{s^2-8s\cancel{+25}-9}e^{-7s}\right\}$. $\rightarrow$ (Type II + Type III).

Sol^n: $F(s) = \frac{6s-4}{s^2-8s\cancel{+25}-9}$

$$\mathcal{L}^{-1}\{F(s)\} = \mathcal{L}^{-1}\left\{\frac{6s-4}{s^2-8s\cancel{+25}-9}\right\}.$$

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Type II $\longrightarrow$ $= \mathcal{L}^{-1}\left\{\frac{6s-4}{(s-4)^2-25}\right\} = e^{4t}\left\{6\cosh(5t)+4\sinh(5t)\right\} = f(t).$

1st Translation

(Type II $\rightarrow$ Ex-2)

$\therefore f(t) = e^{4t}\left(6\cosh(5t)+4\sinh(5t)\right)$

$\therefore f(t-a) = f(t-\pi) = e^{4(t-\pi)}\left(6\cosh(5t-5\pi)+4\sinh(5t-5\pi)\right).$

$\therefore \mathcal{L}^{-1}\left(\frac{6s-4}{s^2-8s-9}e^{-\pi s}\right) = f(t-\pi)\cdot u(t-\pi).$

Ans:

Type-IV (Using partial fraction)

Ex-1: $\mathcal{L}^{-1}\left\{\frac{1}{s(s+1)}\right\}$ [factor in denominator $\rightarrow$ partial fraction].

$\therefore \frac{1}{s(s+1)} = \frac{1}{s(0+1)} + \frac{1}{(s+1)(-1)}$

$= \frac{1}{s} - \frac{1}{s+1}.$

$\frac{1}{s^2+1} = \frac{1}{(s+i)(s-i)}$

cover up rule $\rightarrow$ can't do quadratic func.

$\boxed{\mathcal{L}^{-1}\left\{\frac{1}{s(s^2+1)}\right\}}$

$\therefore \mathcal{L}^{-1}\left\{\frac{1}{s(s+1)}\right\} = \mathcal{L}^{-1}\left\{\frac{1}{s}-\frac{1}{s+1}\right\}$

$= \mathcal{L}^{-1}\left\{\frac{1}{s}\right\} - \mathcal{L}^{-1}\left\{\frac{1}{s-(-1)}\right\}$

$= 1 - e^{-t}.$

Ans:

(H.W)

1. $\mathcal{L}^{-1}\left\{\frac{-s}{(s^2+1)(s+1)}\right\}. \rightarrow A=-\frac{1}{2},\ B=-\frac{1}{2},\ C=\frac{1}{2}$

$\therefore -\frac{s}{(s^2+1)(s+1)} = \frac{As+B}{s^2+1} + \frac{C}{s+1}$

2. $\mathcal{L}^{-1}\left\{\frac{-s}{(s^2+1)(s+1)}e^{-\pi s}\right\}.$

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Ex-2: $\mathcal{L}^{-1}\left\{\dfrac{4s^2-5s}{(s+1)(s-2)^2}\right\}$.

repeated factor → repeated cover up rule.

Soln:

$$F(s) = \frac{4s^2-5s}{(s+1)(s-2)^2}$$

$$= \frac{1}{(s-2)}\left(\frac{4s^2-5s}{(s+1)(s-2)}\right)$$

$$= \frac{1}{(s-2)}\left(\frac{4(-1)^2-5(-1)}{(s+1)(-1-2)} + \frac{4(2)^2-5(2)}{(s-2)(2+1)}\right)$$

$$= \frac{1}{(s-2)}\left(\frac{9}{-3(s+1)} + \frac{6}{3(s-2)}\right)$$

$$= \frac{1}{(s-2)}\left(\frac{-3}{s+1} + \frac{2}{s-2}\right).$$

$$= \frac{-3}{(s-2)(s+1)} + \frac{2}{(s-2)(s-2)}.$$

$$= -3\left(\frac{1}{(s+1)(s-2)}\right) + 2/(s-2)^2$$

$$= -3\left(\frac{1}{(s+1)(-1-2)} + \frac{1}{(s-2)(2+1)}\right) + 2/(s-2)^2$$

$$= -3\left(\frac{1}{-3(s+1)} + \frac{1}{3(s-2)}\right) + \frac{2}{(s-2)^2}$$

$$= \frac{1}{s+1} - \frac{1}{s-2} + \frac{2}{(s-2)^2}$$

$\mathcal{L}^{-1}\left\{\dfrac{1}{(s-2)^2}\right\}$

$F(s) = \dfrac{1}{(s-2)^2}$; $F(s+2) = \dfrac{1}{(s+2-2)^2} = \dfrac{1}{s^2}$

$\mathcal{L}^{-1}(F(s+2)) = t$

$\therefore \mathcal{L}^{-1}\left\{\dfrac{1}{s-2}\right\} = e^{2t}\cdot t$

$$\therefore \mathcal{L}^{-1}\left\{\frac{4s^2-5s}{(s+1)(s-2)^2}\right\} = \mathcal{L}^{-1}\left\{\frac{1}{s+1}\right\} - \mathcal{L}^{-1}\left\{\frac{1}{s-2}\right\} + 2\cdot\mathcal{L}^{-1}\left\{\frac{1}{(s-2)^2}\right\}$$

$$= e^{-t} - e^{2t} + t\cdot e^{2t}.$$ Ans:

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